

ADNI Knowledge Graph: Current State Analysis and Causality Integration Strategy

Progress Report for Patient-Centric Alzheimer's Disease Research Platform

Master's Thesis Project

Repository: github.com/theFellandes/ADNIKnowledgeGraph

January 2025 — Pre-Meeting Discussion Document

Abstract

This report presents a comprehensive analysis of the current state of the ADNI (Alzheimer's Disease Neuroimaging Initiative) Knowledge Graph project, building upon the knowledge-driven multi-tier architecture established in our IEEE Big Data 2025 paper [1]. The system currently operates as a Labeled Property Graph (LPG) in Neo4j with approximately 407,000 nodes and 1.16 million relationships, integrated with Elasticsearch for fast retrieval and a multi-tier image storage system ensuring lossless preservation. This document examines the exploratory data analysis findings, discusses the architectural distinction between LPGs and true semantic Knowledge Graphs, and critically evaluates the path forward for causality integration. We raise key questions regarding ontology customization for causal reasoning—topics intended for discussion in the upcoming meeting. Importantly, we clarify that the temporal graph component serves patient health trajectory tracking rather than being the primary research focus, and that causality analysis represents the ultimate goal of the thesis research plan.

I. INTRODUCTION

The Alzheimer's Disease Neuroimaging Initiative (ADNI) represents one of the most comprehensive longitudinal studies in neurodegenerative disease research. As documented in our IEEE Big Data 2025 paper [1], ADNI generates tens of thousands of high-resolution MRI and PET scans together with clinical, biomarker, and genetic data across multiple study phases (ADNI-1, ADNI-GO, ADNI-2, ADNI-3, ADNI-4). This multi-terabyte data accumulation offers tremendous opportunities for understanding complex neurodegenerative disorders but simultaneously creates severe bottlenecks for existing storage and retrieval infrastructures.

Our previous work addressed the fundamental challenge of integrating semantic modeling through knowledge graphs, high-performance indexing, and multi-tier lossless storage into a single coherent architecture. The system demonstrated that medical images can be managed not only as static files but as semantically enriched nodes within a knowledge network, providing both clinical fidelity and research flexibility under a unified framework. This current report builds upon that foundation to address the next phase of the thesis: the transition from a Labeled Property Graph to a true semantic Knowledge Graph with causality support.

A. Thesis Research Plan

The project follows a four-phase research plan:

1. **Phase 1 (Complete):** Construct a data-driven Knowledge Graph from ADNI data
2. **Phase 2 (Current):** Develop and compare analytical algorithms
3. **Phase 3:** Identify suitable algorithms for specific research questions
4. **Phase 4:** Define experiment protocols with causality analysis as the ultimate goal

We are currently at the transition point between Phase 1 and Phase 2. The LPG infrastructure is operational and validated, and we must now address fundamental questions about semantic enhancement and causality integration before proceeding.

B. Scope Clarification: Patient-Centric Focus

A critical distinction in our approach: this project emphasizes *patient health-life cycles* rather than disease causality per se. The focus is on tracking individual patient trajectories through time, including past visits, medication history, and biomarker evolution. Disease causality (e.g., "Does amyloid deposition cause tau pathology?") remains a downstream research question that depends on first establishing robust temporal patient profiles.

Important Clarification: Temporal Graph Purpose

The temporal graph component is designed for **patient sickness progress tracking**—monitoring how individual patients change over time through their clinical visits. The visit intervals and temporal ordering serve this tracking purpose; they are *not* the main focal point of the research but rather the infrastructure that enables future causal analysis.

II. FOUNDATION: IEEE BIG DATA 2025 ARCHITECTURE

The current implementation builds directly upon the knowledge-driven, multi-tier architecture presented in our IEEE Big Data 2025 paper [1]. This section summarizes the established foundation before discussing the semantic enhancement required for true Knowledge Graph capabilities.

A. Architecture Overview

The system consists of three complementary components that work together to provide semantic integration, high-performance retrieval, and lossless image preservation. Figure 1 illustrates the overall system architecture and data flow.

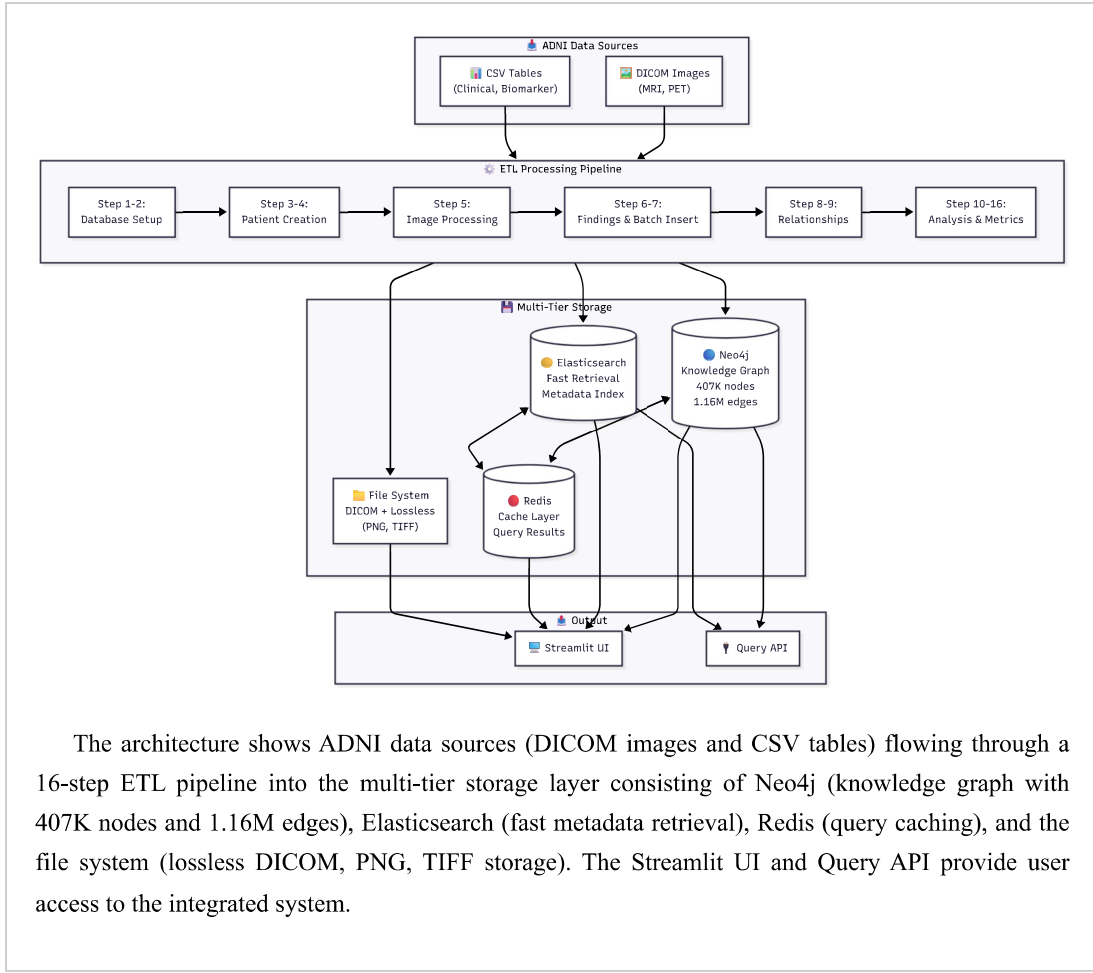


Figure 1: Multi-tier system architecture showing data flow from ADNI sources through the ETL pipeline to storage and retrieval layers.

Neo4j-based Knowledge Graph: A labeled property graph that organizes patients, visits, diagnoses, images, and biomarkers, enriched with ontological annotations. This allows researchers to retrieve complex cross-domain queries such as "all PET scans with a given biomarker along with the subsequent MRI results for the same patient" using a single query.

Elasticsearch-based Indexing: Metadata extracted from DICOM headers and clinical records is indexed to deliver millisecond-level query responses, ensuring high performance in large-scale filtering operations.

Multi-tier File System: Original DICOM images are stored alongside lossless derivatives (16-bit PNG and TIFF files), complemented by JPEG thumbnails for web access. This approach ensures both diagnostic fidelity and user-friendly accessibility.

B. Processing Pipeline

The 15-step ETL pipeline processes ADNI data through sequential stages. As documented in [1], the pipeline achieves ingestion rates exceeding 500 images per second while maintaining

verified lossless conversion through pixel-wise validation. The key pipeline steps are summarized in Table I.

TABLE I: Pipeline Step Summary

Step	Component	Function
1-2	Database Setup & Load	Initialize connections, ingest CSV tables
3-4	Patient & Family	Extract patient entities, family relationships
5	Image Processing	Multi-tier DICOM conversion (PNG, TIFF, thumbnails)
6-7	Findings & Insertion	Parse clinical findings, batch insert to Neo4j
8-9	Relationships & Enhancement	Create temporal chains, semantic relationships
10-16	Analysis	Queries, EDA, biomarker analysis, metrics

Summary of the 16-step ETL pipeline for ADNI data processing.

C. Validated Performance Metrics

The IEEE paper established key performance benchmarks that inform our current development. Query performance was evaluated across five complexity classes, with results summarized in Table II. Simple metadata queries execute in under 10 milliseconds, while complex research queries requiring multiple optional matches take several seconds—a trade-off addressed through caching and index optimization.

TABLE II: Query Performance Summary (from IEEE Big Data 2025)

Complexity	N	Mean (ms)	Median (ms)	P95 (ms)	DB Hits
C1 Simple	15	3.1	1.3	10.3	9.5K
C2 Moderate	15	6.4	2.5	23.7	4.9K
C3 Complex	10	33.6	26.1	85.9	9.9K
C4 Research	10	7318.2	7210.5	14953.5	31.6K
C5 Analytical	10	51.4	50.9	106.7	62.5K

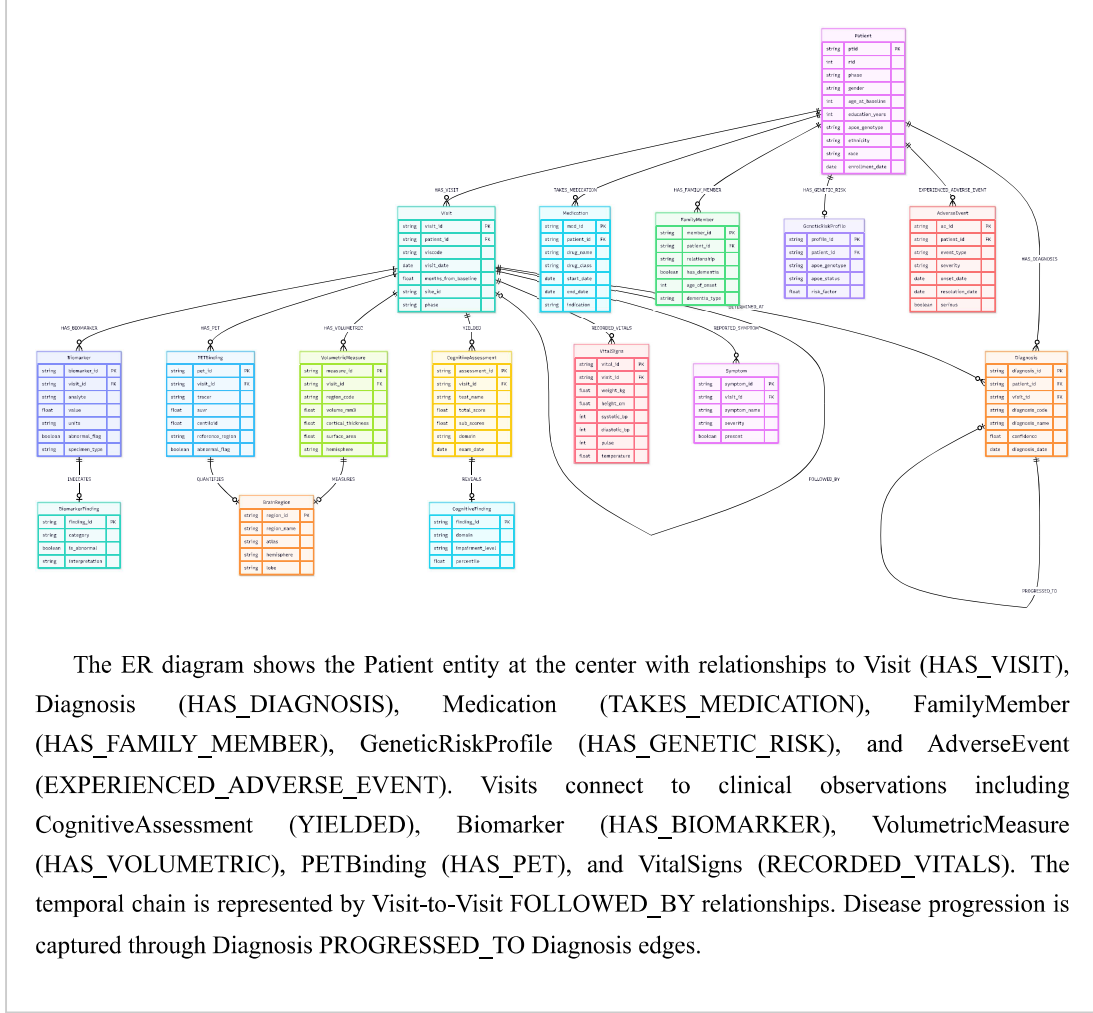
Query performance metrics by complexity class. C4 Research queries are slow due to multiple optional matches; Redis caching mitigates this for frequent queries.

III. EXPLORATORY DATA ANALYSIS: CURRENT GRAPH STATE

Building upon the foundation established in the IEEE paper, this section presents the current state of the knowledge graph through exploratory data analysis. The graph has evolved since the initial publication to include additional relationship types and enhanced node properties.

A. Graph Statistics Overview

The current Neo4j graph database contains the statistics summarized in Table III. These metrics reflect the patient-centric design philosophy where each patient serves as a hub node connected to their visits, diagnoses, biomarkers, images, and genetic profiles. Figure 2 presents the entity-relationship structure of the current LPG implementation.



The ER diagram shows the Patient entity at the center with relationships to Visit (HAS_VISIT), Diagnosis (HAS_DIAGNOSIS), Medication (TAKES_MEDICATION), FamilyMember (HAS_FAMILY_MEMBER), GeneticRiskProfile (HAS_GENETIC_RISK), and AdverseEvent (EXPERIENCED_ADVERSE_EVENT). Visits connect to clinical observations including CognitiveAssessment (YIELDED), Biomarker (HAS_BIOMARKER), VolumetricMeasure (HAS_VOLUMETRIC), PETBinding (HAS_PET), and VitalSigns (RECORDED_VITALS). The temporal chain is represented by Visit-to-Visit FOLLOWED_BY relationships. Disease progression is captured through Diagnosis PROGRESSED_TO Diagnosis edges.

Figure 2: Entity-Relationship diagram of the current Labeled Property Graph implementation in Neo4j, showing the patient-centric design with all node types and relationship patterns.

TABLE III: Current Database Statistics

Entity	Count	Description
Patients	2,638	Unique ADNI participants across all phases
Visits	30,267	Clinical visits with temporal ordering
Diagnoses	25,946	Diagnostic assessments (CN, MCI, AD, etc.)
Total Nodes	~407,000	All entity types including images, biomarkers
Total Relationships	~1,160,000	Edges connecting entities

B. Node Type Distribution

The graph contains diverse node types representing different aspects of patient data. Image nodes dominate the count due to the multi-modal imaging nature of ADNI (MRI and PET scans at each visit). The distribution reflects the comprehensive nature of the dataset, with nodes representing clinical assessments, biomarker measurements, genetic profiles, and family history information.

Analysis reveals that each patient has an average of approximately 11.5 visits over the study duration, with each visit potentially generating multiple imaging studies, cognitive assessments, and biomarker measurements. This creates a rich, interconnected structure suitable for longitudinal trajectory analysis.

C. Relationship Pattern Analysis

The relationship structure reflects the patient-centric design. Figure 3 illustrates a detailed view of a single patient node with all connected entities and their properties, demonstrating the richness of the data model.

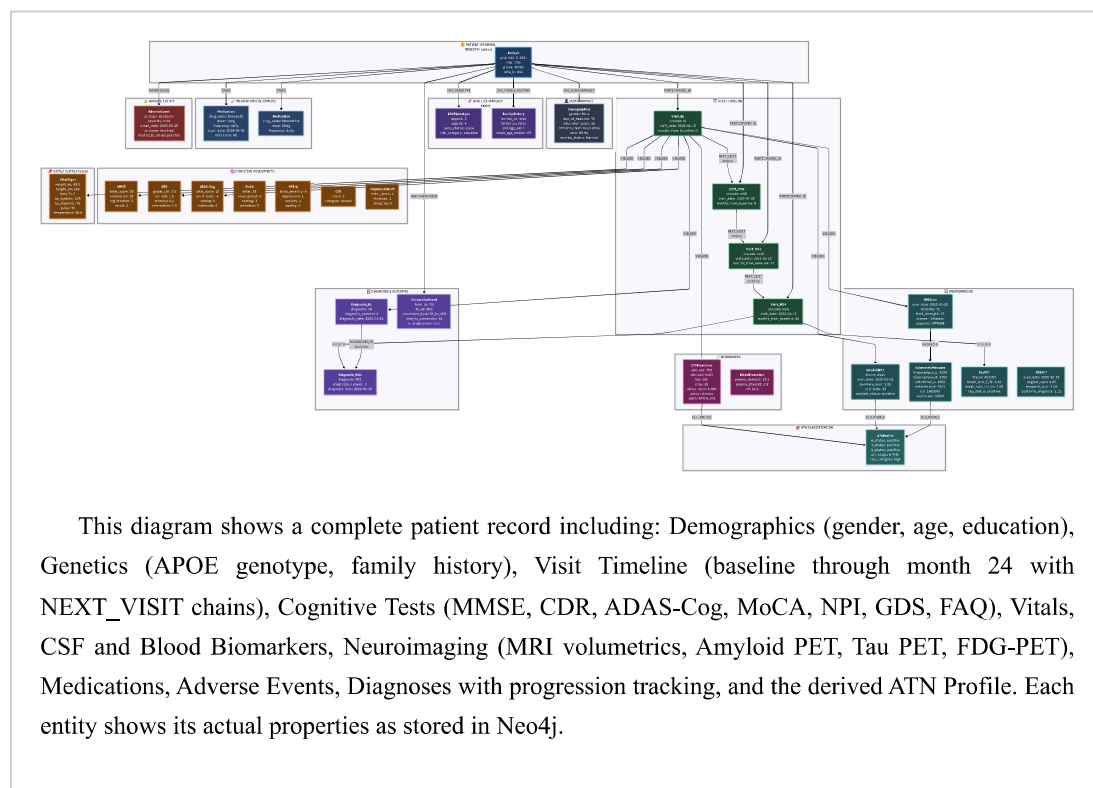


Figure 3: Detailed schema of a single patient node showing all connected entities, their properties, and relationships. This illustrates the comprehensive data capture for each ADNI participant.

Key relationship types include:

- **HAS_VISIT (~30,000):** Links patients to their clinical visits
- **FOLLOWED_BY (~27,000):** Creates temporal chains between consecutive visits
- **HAS_DIAGNOSIS (~26,000):** Associates visits with clinical diagnoses
- **HAS_BIOMARKER (~45,000):** Links visits to CSF and blood biomarker measurements
- **HAS_IMAGE (~150,000):** Connects visits to imaging studies
- **PROGRESSED_TO (~2,500):** Captures disease stage transitions

D. Disease Stage Distribution

The ADNI cohort spans the full Alzheimer's disease continuum, enabling analysis across all disease stages. Table IV shows the distribution of patients by their most recent diagnosis, demonstrating balanced representation across the clinical spectrum.

TABLE IV: Patient Distribution by Disease Stage

Stage	Description	Approx. Count	Percentage
CN	Cognitively Normal	650	24.6%
SMC	Subjective Memory Concern	150	5.7%
EMCI	Early Mild Cognitive Impairment	520	19.7%
LMCI	Late Mild Cognitive Impairment	680	25.8%
AD	Alzheimer's Disease Dementia	638	24.2%

Distribution of patients across disease stages at most recent visit.

E. Patient Connectivity Analysis

Analysis of patient node connectivity reveals the richness of the graph structure. The average patient node has connections to 11-12 visit nodes, each visit connecting to multiple clinical observations. High-connectivity patients (those with longer study participation) may have over 100 direct and indirect relationships, demonstrating the value of the graph model for capturing longitudinal complexity.

Super-connector analysis identifies patients with the most comprehensive data profiles—these individuals often have complete multi-modal data (MRI, PET, CSF, genetics) across many visits and represent ideal candidates for detailed trajectory analysis.

IV. CRITICAL DISTINCTION: LPG VS. TRUE KNOWLEDGE GRAPH

A fundamental issue must be addressed before proceeding with causality integration: the current Neo4j implementation is a *Labeled Property Graph (LPG)*, not a true semantic Knowledge Graph. This distinction has significant implications for reasoning capabilities and is often misunderstood in the literature.

A. Defining the Distinction

Many systems incorrectly claim to be Knowledge Graphs when they are actually LPGs. The key differentiator is *semantic grounding*—whether relationships carry formal, machine-interpretable meaning that enables reasoning and inference. Figure 4 illustrates this fundamental distinction.

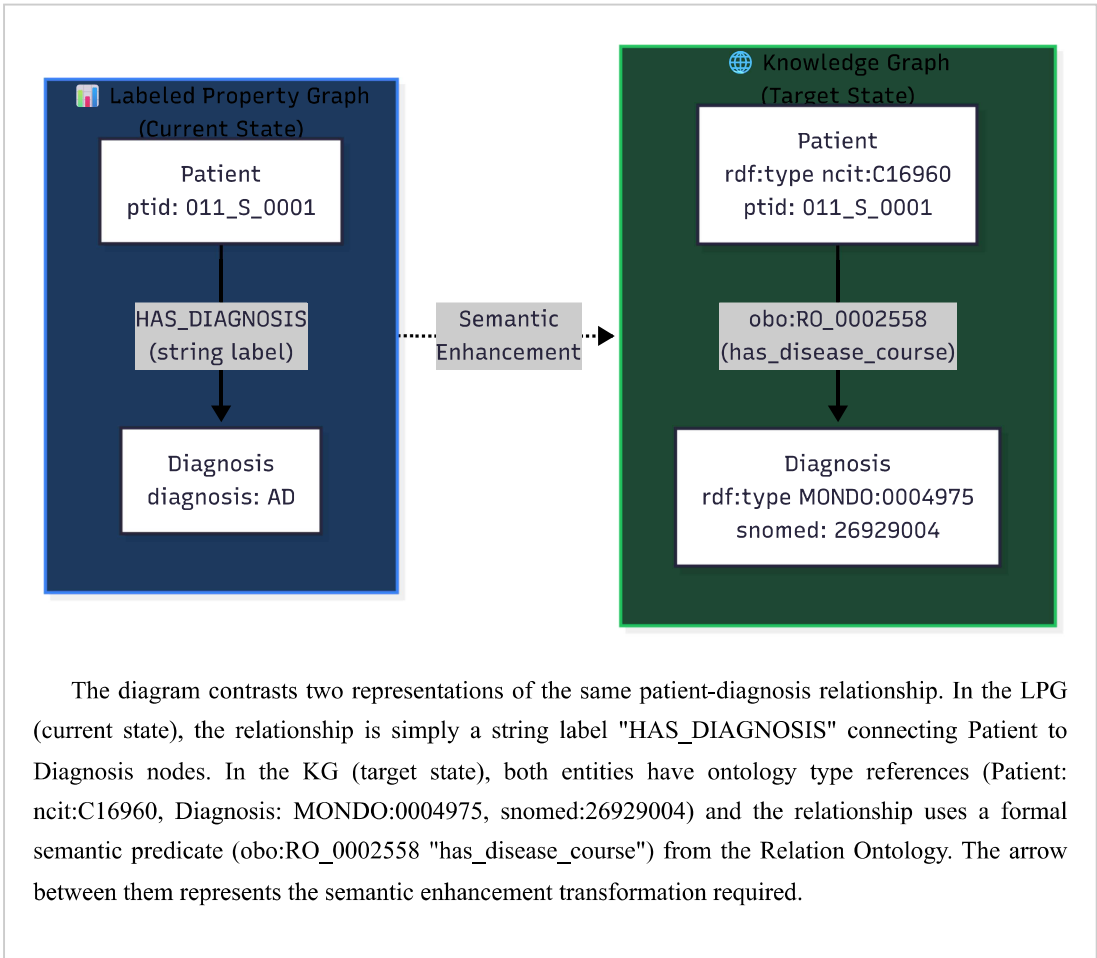


Figure 4: Side-by-side comparison of LPG and Knowledge Graph representations. The LPG uses string labels while the KG uses ontology-grounded URIs that enable reasoning.

TABLE V: LPG vs. Knowledge Graph Characteristics

Aspect	LPG (Current State)	True KG (Target State)
Schema	Implicit, application-defined	Formal OWL/RDF ontology
Relationships	Typed strings (syntactic)	Ontology-grounded (semantic)
Inference	Not supported	Reasoning capabilities
Interoperability	System-specific	Standard ontologies (SNOMED, LOINC)
Query Language	Cypher only	SPARQL + Cypher

B. Current LPG Limitations

In the current implementation, a relationship like `HAS_DIAGNOSIS` is simply a string label connecting two nodes. It does not carry formal semantic content that a reasoner could interpret. For example, the system cannot automatically infer that a patient with diagnosis "AD" has a more severe condition than one with "MCI" without explicit programming of this logic.

Similarly, biomarker values are stored as properties without semantic links to what they measure. The system knows that a patient has an "ABETA42" value of 850, but it does not know that this measures amyloid-beta 42 peptide in cerebrospinal fluid, that values below 880 pg/mL are typically considered abnormal, or that this relates to amyloid plaque pathology in the brain.

C. Semantic Enhancement Strategy

The transformation from LPG to true KG requires adding ontology references and semantic relationship grounding. Figure 5 presents the target semantic architecture with full ontology integration, and Table VI outlines the planned ontology mappings.

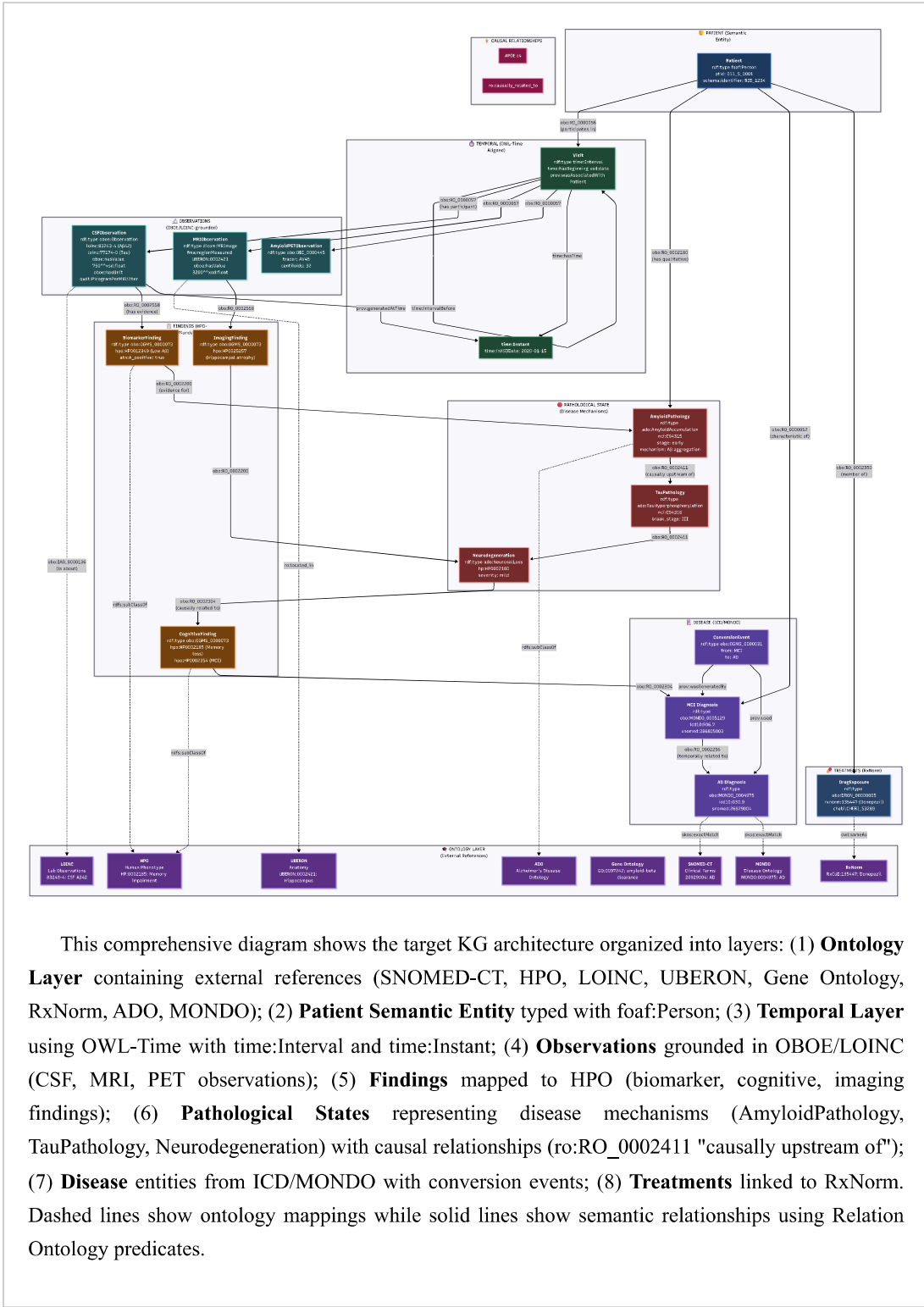


Figure 5: Target semantic Knowledge Graph architecture showing ontology layer integration, semantic relationship predicates from RO, and the causal cascade representation ($A \rightarrow T \rightarrow N$).

TABLE VI: Planned Ontology Mappings

Entity Type	Target Ontology	Example Mapping
Diagnosis	SNOMED-CT	AD → SNOMED:26929004
Biomarker	LOINC	Aβ42 → LOINC:72333-6
Brain Region	UBERON	Hippocampus → UBERON:0002421
Symptom	HPO	Memory loss → HP:0002354
Relationship	RO (Relation Ontology)	HAS_DIAGNOSIS → RO:0000091

Ontology mapping strategy for semantic enhancement.

Figure 6 shows the domain-organized view of the enhanced Knowledge Graph, highlighting how different data domains (Clinical, Biomarker, Imaging, Genetic) connect through the patient-centric model with semantic relationships.

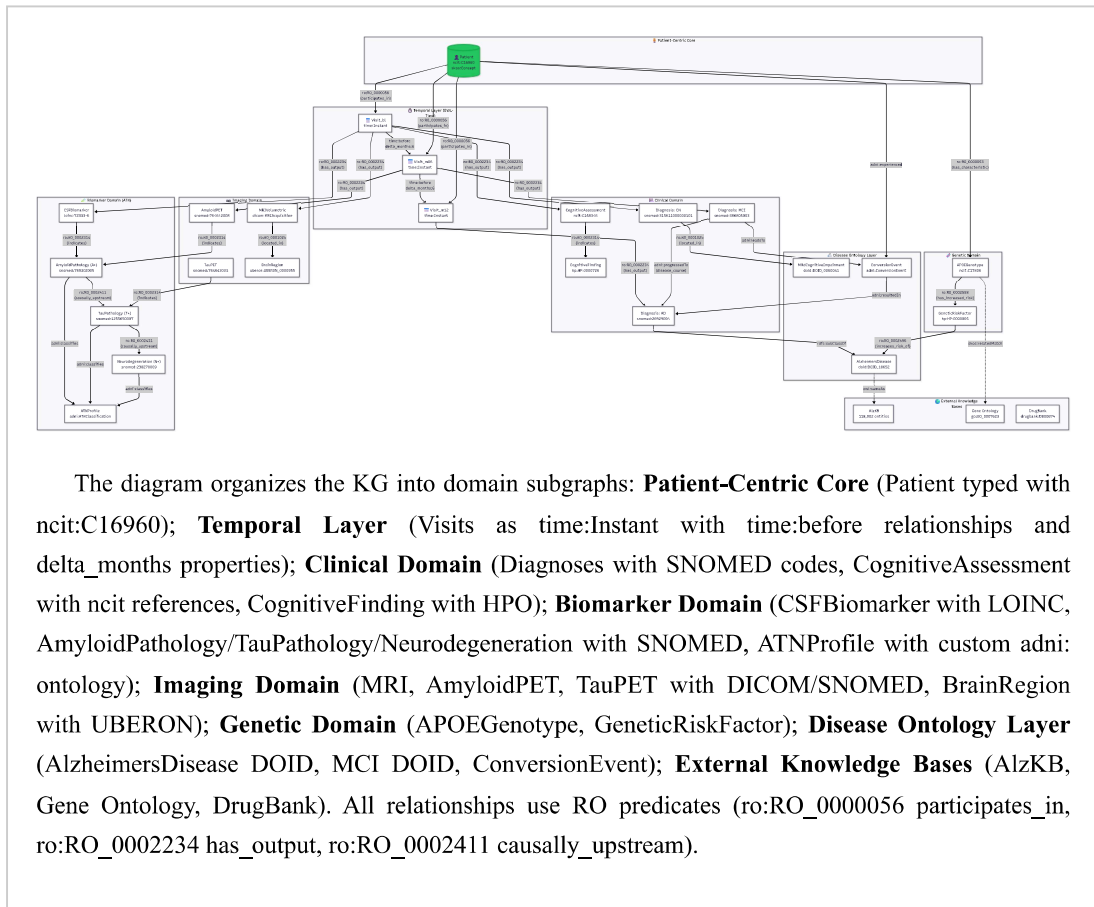


Figure 6: Domain-organized Knowledge Graph showing semantic relationships between Patient, Clinical, Biomarker, Imaging, and Genetic domains with Relation Ontology predicates.

D. Why This Matters for Causality

True causal reasoning requires semantic grounding. Without formal ontologies, we cannot distinguish between correlation and causation at the data model level. For example:

- **Correlation (current capability):** "Patients with low A β 42 often have memory problems"
- **Causation (requires semantic model):** "Amyloid deposition mechanistically leads to synaptic dysfunction"

Ontologies provide the semantic framework needed to model causal mechanisms, not just statistical associations. This is critical for the thesis goal of causality analysis.

V. TEMPORAL GRAPH: ROLE AND SCOPE

The temporal graph component requires clarification regarding its purpose within the overall research plan. There has been some ambiguity about whether temporal analysis is a primary research focus; this section clarifies its role.

A. Primary Purpose: Patient Trajectory Tracking

The temporal graph exists to **track patient sickness progression**—following individual patients through their clinical journey over time. The `FOLLOWED_BY` relationship chains establish temporal ordering between visits, while `delta_months` properties capture the intervals between observations. Figure 7 illustrates a typical patient trajectory showing disease progression over time.

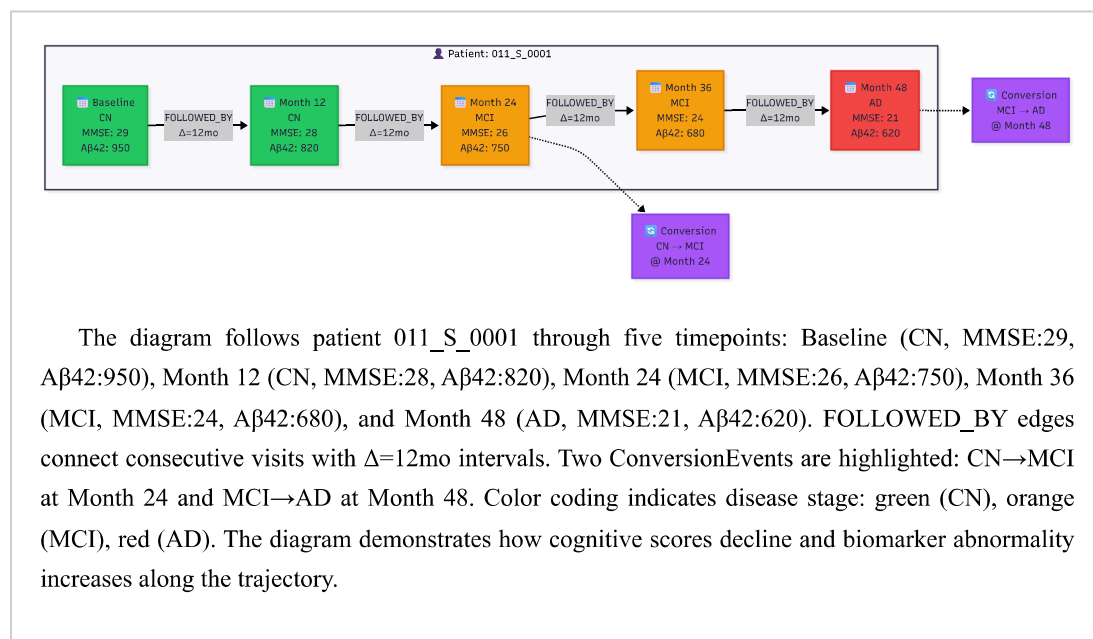


Figure 7: Example patient trajectory showing disease progression from Cognitively Normal (CN) through MCI to AD over 48 months, with biomarker and cognitive score changes at each visit.

This infrastructure enables queries such as:

- What was this patient's cognitive trajectory over 5 years?
- When did biomarker abnormalities first appear relative to clinical symptoms?

- How long did it take for this patient to progress from MCI to AD?

B. What Temporal Graph Provides

- **Visit Ordering:** FOLLOWED_BY chains establish temporal sequence
- **Time Deltas:** Properties capture intervals between observations
- **Baseline Reference:** `months_from_baseline` provides consistent timeline
- **Conversion Events:** Track when patients transition between disease stages

C. What Temporal Graph Does NOT Provide

It is important to distinguish what the temporal infrastructure does not provide, as these represent the gap to be addressed in future phases:

- Causal relationships between biomarkers and outcomes
- Mechanistic understanding of disease progression
- Inference about interventional effects
- Counterfactual reasoning capabilities

The temporal ordering is *necessary but not sufficient* for causal analysis. Temporal precedence (A occurs before B) does not establish causation (A causes B). The transition from temporal description to causal explanation is a core challenge for the thesis.

VI. CAUSALITY: CURRENT STATE AND OPEN QUESTIONS

We have **not yet explored the causality part** of the thesis research plan. This section documents the current state and raises questions for discussion before proceeding.

A. Current Capabilities

The system currently supports:

- ✓ Temporal ordering of observations
- ✓ Longitudinal patient trajectories
- ✓ PROGRESSED_TO edges for stage transitions
- X No formal causal semantics
- X No causal discovery algorithms implemented
- X No intervention modeling

The ATN (Amyloid-Tau-Neurodegeneration) framework provides a biological staging system that could inform causal modeling. Figure 8 illustrates how biomarker inputs are processed to generate ATN classifications.

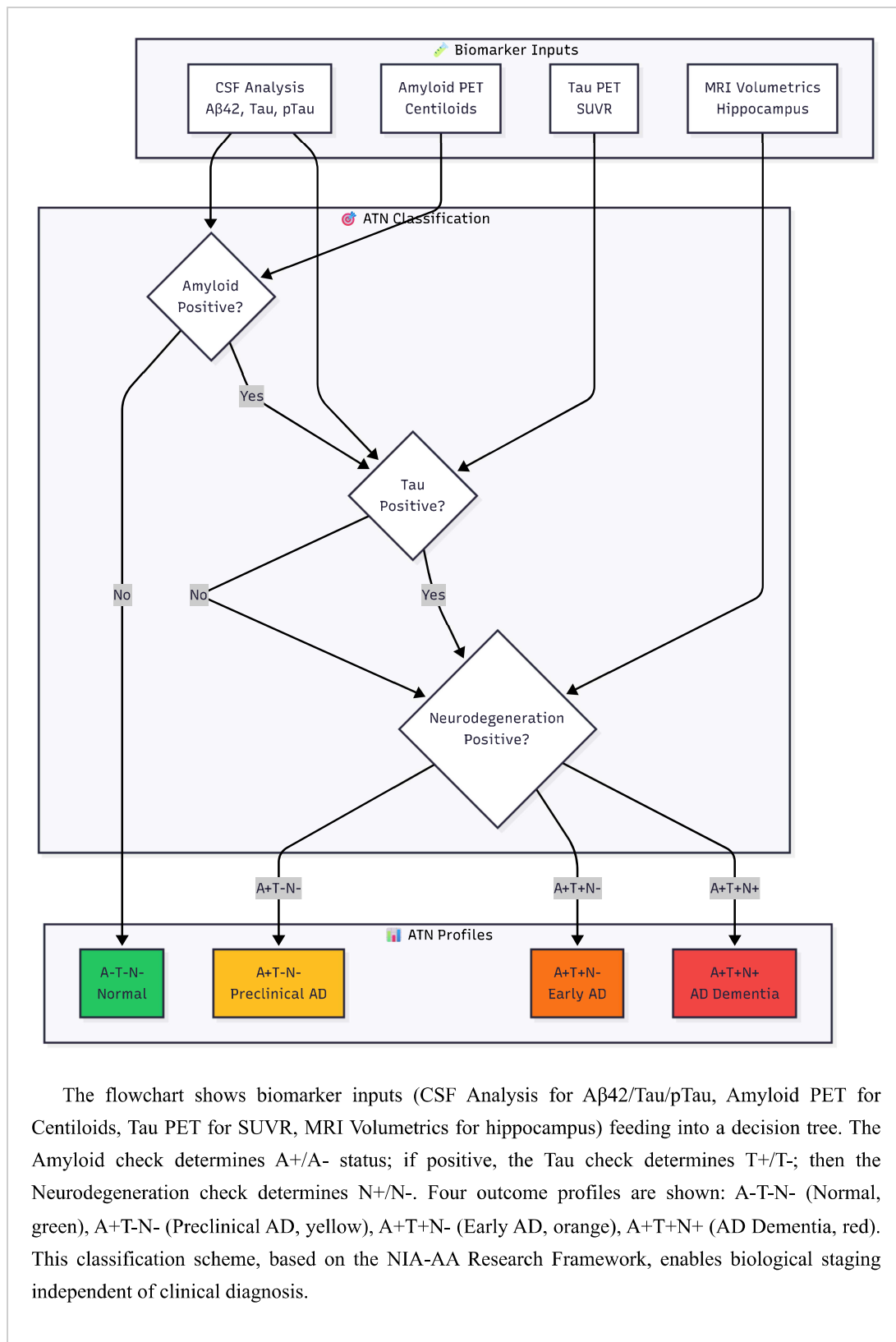


Figure 8: ATN classification decision flow showing how CSF biomarkers, PET imaging, and MRI volumetrics combine to determine amyloid (A), tau (T), and neurodegeneration (N) status.

B. Planned Causal Discovery Algorithms

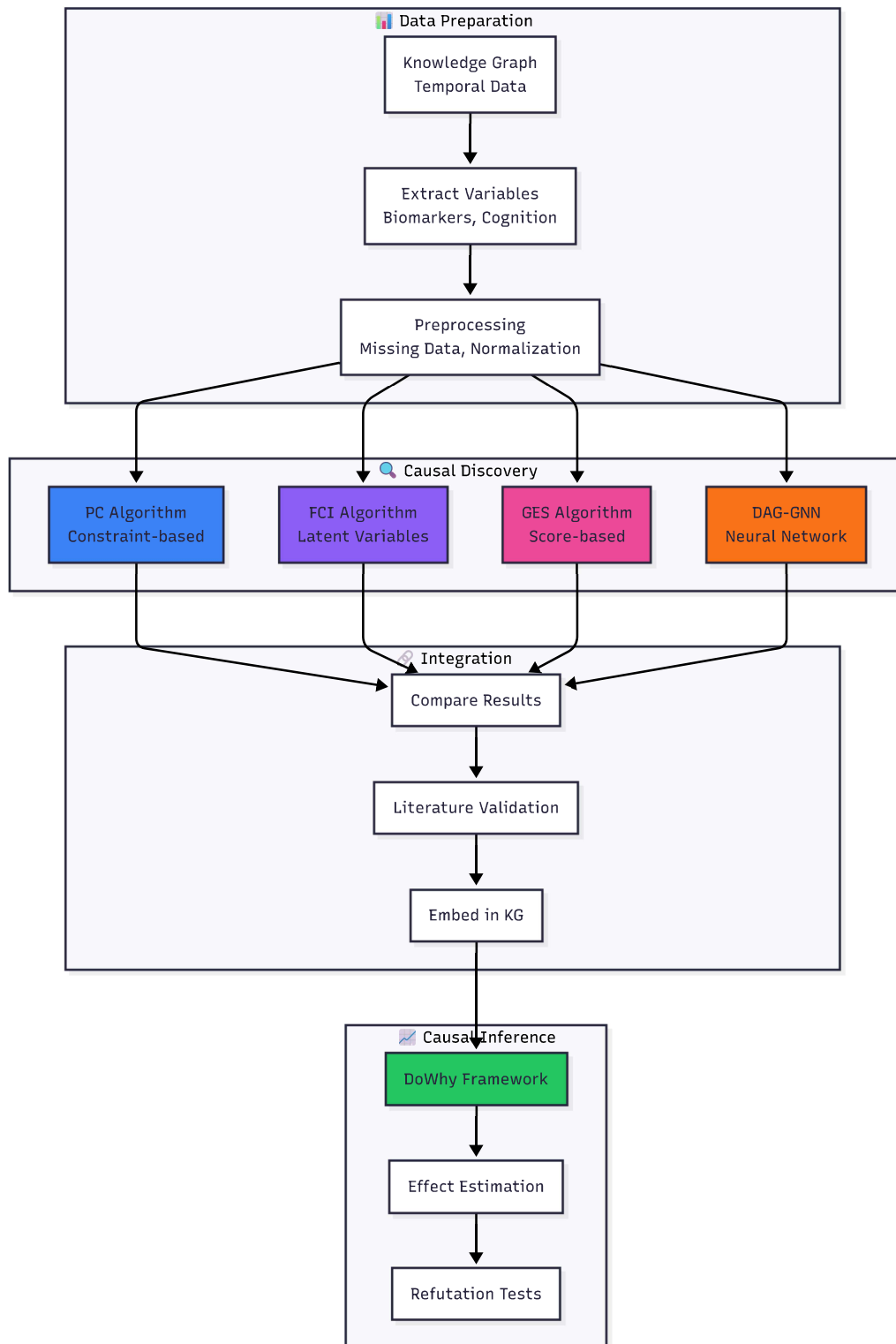
The thesis plan includes evaluation of several causal discovery approaches, each with different assumptions and capabilities. Table VII summarizes the planned algorithms.

TABLE VII: Planned Causal Discovery Algorithms

Algorithm	Type	Strengths	Limitations
PC Algorithm	Constraint-based	Well-established, interpretable	Assumes causal sufficiency
FCI Algorithm	Constraint-based	Handles latent confounders	Computationally expensive
GES	Score-based	Good with sparse graphs	Score function sensitivity
DAG-GNN	Neural network	Scalable, nonlinear	Less interpretable
DoWhy	Inference framework	Refutation tests, robust	Requires prior DAG

Summary of causal discovery algorithms planned for Phase 4.

Figure 9 presents the planned causal discovery pipeline, showing how data flows from the Knowledge Graph through multiple discovery algorithms to integration and inference.



The pipeline consists of four stages: (1) **Data Preparation**—extracting variables (biomarkers, cognition) from the Knowledge Graph and preprocessing (missing data handling, normalization); (2) **Causal Discovery**—running multiple algorithms in parallel: PC Algorithm (constraint-based), FCI Algorithm (handles latent variables), GES Algorithm (score-based), and DAG-GNN (neural network approach); (3) **Integration**—comparing results across algorithms, validating against literature, and

embedding discovered causal edges into the KG; (4) **Causal Inference**—using the DoWhy framework for effect estimation and refutation tests. This multi-algorithm approach allows comparison of discovered structures and increases confidence in robust causal relationships.

Figure 9: Planned causal discovery pipeline showing data preparation, parallel algorithm execution (PC, FCI, GES, DAG-GNN), integration with literature validation, and DoWhy-based causal inference.

C. The Ontology Question for Causality

A fundamental question emerges: **Should we customize ontologies specifically to support causal reasoning?** This question has no obvious answer and requires discussion before implementation proceeds.

Three approaches are possible:

1. **Option A:** Use standard biomedical ontologies (SNOMED-CT, LOINC, RO) without modification, and layer causal analysis on top as a separate analytical process.
2. **Option B:** Extend standard ontologies with custom causal predicates (CAUSES, PREVENTS, CONFOUNDS) embedded in the graph structure.
3. **Option C:** Create a separate "causal overlay" ontology that references standard concepts but adds causal semantics without modifying the base ontologies.

Each option involves trade-offs between simplicity, expressivity, and standards compliance that should be discussed before implementation.

VII. QUESTIONS FOR DISCUSSION

The following questions are raised for discussion in the upcoming meeting. These represent critical decision points that will shape the next phase of the thesis.

Q1: Ontology Customization for Causality

Should we customize ontologies specifically to reveal and support causal reasoning? The options range from using standard ontologies as-is (simpler but potentially limiting) to creating custom causal extensions (more complex but more expressive). What approach best serves the thesis objectives?

Q2: Approach to Causality

How should we approach the causality component? Options include: (a) Focus on causal discovery from observational data (PC, FCI, GES); (b) Focus on causal inference given an assumed DAG (DoWhy); (c) Combine discovery and inference in a two-stage pipeline; (d) Use knowledge graph structure to guide causal hypothesis generation. Each has different data requirements and scientific contributions.

Q3: Knowledge Graph vs. Causal Graph Relationship

What is the relationship between the semantic KG and causal DAG? Should the causal DAG be embedded within the KG as special edge types? Should it be a separate structure derived from KG data? How do ontological relationships (IS_A, PART_OF) relate to causal relationships?

Q4: Temporal vs. Causal Precedence

How should temporal ordering inform causal inference? Temporal precedence is necessary but not sufficient for causation. Should FOLLOWED_BY edges constrain causal graph learning? How do we handle biomarker changes that precede clinical symptoms by years?

Q5: ATN Framework Integration

Should the ATN (Amyloid-Tau-Neurodegeneration) classification drive causal modeling? The ATN provides a biological framework for AD staging, and the $A \rightarrow T \rightarrow N$ cascade is a hypothesized causal chain. Should we test/validate this cascade using causal discovery methods, or use it as prior knowledge to constrain discovery?

Q6: External Knowledge Base Integration

Should we integrate external knowledge bases (e.g., AlzKB with 118,902 entities) before or after causal analysis? Integration could provide prior knowledge for causal discovery, but risks biasing discovery toward known relationships rather than novel findings.

VIII. PROPOSED NEXT STEPS

A. Immediate Actions (January 2025)

1. Complete LPG documentation for thesis methodology chapter
2. Finalize ontology mapping strategy based on discussion outcomes

3. Generate comprehensive EDA statistics and visualizations

B. Short-term Goals (Q1 2025)

1. Implement ontology reference properties (SNOMED, LOINC codes)
2. Create ATNProfile nodes and classification system
3. Evaluate first causal discovery algorithm (PC or FCI)
4. Design experiment protocol for algorithm comparison

C. Medium-term Goals (Q2 2025)

1. Complete semantic enhancement (LPG $\rightarrow$ KG transformation)
2. Integrate AlzKB for biological context enrichment
3. Implement multiple causal discovery algorithms
4. Develop causal inference framework with DoWhy

IX. CONCLUSION

This report has presented the current state of the ADNI Knowledge Graph project, building upon the foundation established in our IEEE Big Data 2025 paper [1]. The main findings can be summarized as follows:

1. **Infrastructure Validated:** The 16-step pipeline successfully processes ADNI data into a Neo4j graph with approximately 407,000 nodes and 1.16 million relationships, with query performance and lossless image handling validated in the IEEE paper.
2. **LPG, Not KG:** The current implementation is a Labeled Property Graph; semantic enhancement through ontology mapping is required for true Knowledge Graph capabilities that support reasoning.
3. **Temporal Tracking Operational:** FOLLOWED_BY chains and visit ordering support patient trajectory analysis, which serves as infrastructure for—not the focus of—the research.
4. **Causality Unexplored:** Causal discovery and inference remain future work, pending decisions on ontology customization and algorithmic approach.
5. **Key Questions Raised:** Decisions on ontology customization, causal graph integration, and ATN framework validation are needed before proceeding to Phase 2.

The questions raised in Section VII are intended to guide discussion in the upcoming meeting, ensuring alignment between the technical implementation and thesis objectives. The causality analysis represents the ultimate goal of the research plan, and decisions made now will shape the path toward that goal.

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APPENDIX A: ADNI DATA SCHEMA SUMMARY

The ADNI dataset (headers.json) contains 180+ tables with 5,800+ unique columns covering clinical, imaging, biomarker, and genetic domains. Key table categories include:

TABLE A-I: ADNI Data Schema Categories

Category	Example Tables	Key Fields
Clinical	DXSUM, ADAS, CDR, MMSE	DIAGNOSIS, TOTSCORE, CDRSUM
Biomarker	UPENNBBIOMK, CSF_BIOMARKERS	ABETA42, TAU, PTAU
Imaging	All_Images, AMYMETA, TAUMETA	image_id, modality, SCANDATE
Genetics	APOERES, GENETICS	APOE genotype, SNP data
Demographics	PTDEMOG, VITALS	AGE, EDUCATION, GENDER

Summary of ADNI data categories available for knowledge graph construction.

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